Information Theory Course

Entropy, Relative Entropy, and Mutual Information

Lecture notes
based on Chapter 2 of Cover and Thomas [2006]
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Draft Lecture notes, version of Aug 17

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1 Entropy

Definition 1.1 (Entropy)**.** If X has probability mass function p_X on a finite or countable alphabet $\mathcal{X}$, the *entropy* of X is

$$H(X) \;=\; H(p_X) \;=\; -\sum_{x\in\mathcal{X}} p_X(x)\log p_X(x).$$

Entropy depends on X only through its law p_X, not on the numerical or semantic labels attached to the outcomes: relabelling the alphabet by a bijection leaves $H(X)$ unchanged.

This is why theorem 1.1 is written equivalently as $H(X)$ or $H(p_X)$, or simply $H(p)$ when the random variable is understood.

Remark 1.2 (Entropy as expected surprise). Write $s(x) = \log\big(1/p_X(x)\big)$ for the *surprise* of an outcome x with $p_X(x) > 0$: an outcome that was already almost certain ($p_X(x) \approx 1$) carries little surprise, while a rare outcome ($p_X(x) \approx 0$) carries a lot. Entropy is exactly the expected surprise,

$$H(X) \;=\; \mathbb{E}\left[\log \frac{1}{p_X(X)}\right],$$

which is Cover and Thomas [2006, eq. (2.3)]. Cover and Thomas call this the "eerily self-referential" expectation of $\log\big(1/p(X)\big)$ under $p(x)$ itself: the same distribution both generates X and measures how surprising each value of X is.

Proposition 1.3 (Elementary properties)**.** *For a finite-valued random variable X on an alphabet of size $m = |\mathcal{X}|$:*

1. $H(X) \geq 0$ *[Cover and Thomas, 2006, Lemma 2.1.1].*
2. $H(X) = 0$ *if and only if X is constant with probability* 1.
3. *If logarithms to bases $a, b > 1$ are used, then $H_b(X) = (\log_b a)\, H_a(X)$ [Cover and Thomas, 2006, Lemma 2.1.2].*
4. $H(X) \leq \log m$*, with equality if and only if X is uniform on $\mathcal{X}$. (Proved in theorem 5.7 below, once we have relative entropy available. It is not a consequence of the first three parts.)*

Proof. Part 1: each term $-p(x)\log p(x) = p(x)\log\big(1/p(x)\big)$ is nonnegative because $0 \leq p(x) \leq 1$ makes $1/p(x) \geq 1$. Part 2: the sum of nonnegative terms is zero exactly when every term is zero, *i.e.* every x with $p(x) > 0$ has $p(x) = 1$, which (since probabilities sum to 1) means there is exactly one such x. Part 3 is the change-of-base identity $\log_b t = (\log_b a)\log_a t$, applied inside the sum. Part 4 is deferred to theorem 5.7. ∎

Example 1.4 (A biased coin, Cover and Thomas, 2006, Example 2.1.1)**.** Let X be Bernoulli(p), taking value 1 with probability p and 0 with probability $1 - p$. Then

$$H(X) \;=\; -p\log p - (1-p)\log(1-p) \;=\; H(p)\,,$$

exactly the binary entropy function of the Conventions above. In particular $H(X) = 1$ bits when $p = \frac{1}{2}$: a fair coin flip carries exactly one bit of uncertainty. Figure 1 plots $H(p)$ against p: it is 0 at the endpoints (X is then not random at all) and reaches its maximum of 1 bits at $p = \frac{1}{2}$, in agreement with theorem 1.3(4) applied to the two-point alphabet $\{0, 1\}$.

Example 1.5 (A dyadic source and an exact question tree)**.** Let X take four values with probabilities

$$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$$

(Cover and Thomas, 2006, Example 2.1.2). Then

$$H(X) = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = \frac{7}{4}\,\text{bits}.$$

There is a binary decision tree with codeword lengths $1, 2, 3, 3$ ("is $X = 1$?", then "is $X = 2$?", then "is $X = 3$?"), so its expected number of yes/no questions is exactly $7/4$. The equality is

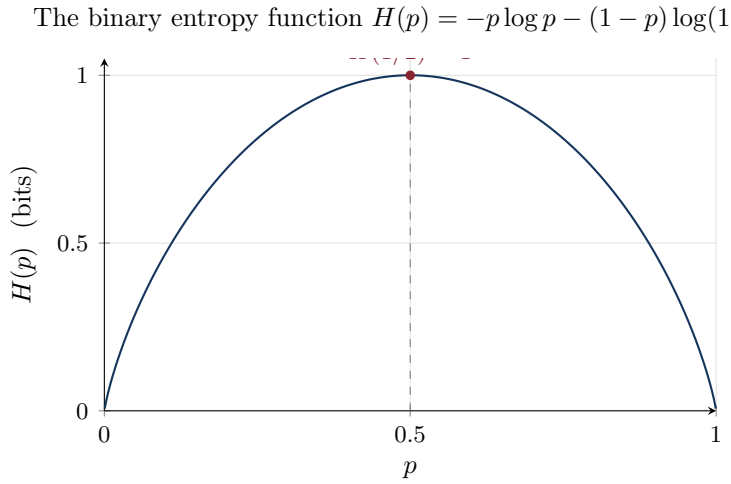


Figure 1: The binary entropy function, after Cover and Thomas [2006, Fig. 2.1]. It is concave, vanishes at $p \in \{0, 1\}$ (no randomness), and is maximised at $p = \frac{1}{2}$ (maximal randomness for a two-valued outcome).

not an accident: probabilities of the form $2^{-\ell}$ fit a complete binary tree exactly. Chapter 5 of Cover and Thomas [2006] develops the general coding theorem, which shows that $H(X)$ is always a lower bound on the expected number of yes/no questions, attained exactly when every probability is a negative power of 2.

Key idea: why log?

Shannon did not simply postulate $H(X) = -\sum p(x)\log p(x)$. He showed in Shannon [1948, Appendix 2] that it is the *only* functional (up to a positive multiplicative constant, which fixes the base of the logarithm) that is continuous in the probabilities, increasing in n for the uniform law on n outcomes, and additive under independence.

2 Joint and conditional entropy

There is nothing conceptually new in extending entropy to a pair of random variables: the pair (X, Y) is itself a single vector-valued random variable, so theorem 1.1 already tells us what to do [Cover and Thomas, 2006, §2.2].

Definition 2.1 (Joint entropy)**.** For a pair (X, Y) with joint mass function p_{XY},

$$H(X,Y) = -\sum_{x,y} p_{XY}(x,y)\log p_{XY}(x,y) = \mathbb{E}\left[\log \frac{1}{p_{XY}(X,Y)}\right].$$

Definition 2.2 (Conditional entropy)**.** The conditional entropy of Y given X is the average, over X, of the entropy of the conditional law of Y:

$$\begin{aligned} H(Y|X) &= \sum_{x} p_X(x) H(Y \mid X = x) \\ &= -\sum_{x,y} p_{XY}(x,y)\log p_{Y|X}(y \mid x) \\ &= \mathbb{E}\left[\log \frac{1}{p_{Y|X}(Y \mid X)}\right]. \end{aligned}$$

Only values x with $p_X(x) > 0$ affect the sum.

Theorem 2.3 (Chain rule for two variables)**.**

$$H(X, Y) = H(X) + H(Y|X)\,.$$

Proof. On the support of p_{XY}, $p_{XY}(x, y) = p_X(x)p_{Y|X}(y \mid x)$. Taking $-\log$ of both sides and averaging under p_{XY},

$$\begin{aligned} H(X, Y) &= \mathbb{E}\left[\log \frac{1}{p_{XY}(X, Y)}\right] \\ &= \mathbb{E}\left[\log \frac{1}{p_X(X)} + \log \frac{1}{p_{Y|X}(Y \mid X)}\right] \\ &= H(X) + H(Y|X)\,. \end{aligned}$$

■

Corollary 2.4.

$$H(X, Y|Z) = H(X|Z) + H(Y|X, Z)\,.$$

Proof. Identical computation, conditioned throughout on $\{Z = z\}$ and then averaged over z. ■

Interchanging the roles of X and Y in theorem 2.3 gives

$$H(X) + H(Y|X) = H(X, Y) = H(Y) + H(X|Y)\,,$$

so $H(X) - H(X|Y) = H(Y) - H(Y|X)$: this common quantity is exactly the mutual information of section 3. On its own, however, $H(Y|X)$ need *not* equal $H(X|Y)$: they are two different weighted averages, and the two worked examples below show this concretely.

Example 2.5 (A random key that removes uncertainty)**.** Let X and K be independent fair bits and put $Y = X \oplus K$, where $\oplus$ is addition modulo 2. Then Y is fair, $X \perp\!\!\!\perp Y$ (this is the one-time pad: without the key, the ciphertext Y reveals nothing about the message X), and

$$H(X) = 1\,\text{bits}, \qquad H(X|Y) = 1\,\text{bits}.$$

If the key K is also revealed, $X = Y \oplus K$, hence

$$H(X|Y, K) = 0.$$

The example separates "seeing a ciphertext" from "seeing a ciphertext and its key" using conditional entropy alone: the former leaves all of X's uncertainty intact, the latter removes it completely.

The book's worked example

Example 2.6 (Cover and Thomas, 2006, Example 2.2.1)**.** Let (X, Y) have the joint distribution

$Y\backslash X$	1	2	3	4	$p(y)$
1	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
2	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{4}$
3	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{16}$	$\frac{1}{4}$
4	$\frac{1}{4}$	0	0	0	$\frac{1}{4}$
$p(x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	1

The marginal of X is $p(x) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$ and that of Y is uniform, $p(y) = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)$, so $H(X) = \frac{7}{4}$ bits (theorem 1.5) and $H(Y) = \log_2 4 = 2$ bits. Cover and Thomas compute

$$\begin{aligned} H(X|Y) &= \sum_{i=1}^{4} p(Y=i)\, H(X \mid Y=i) \\ &= \tfrac{1}{4} H\left(\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{8}\right) + \tfrac{1}{4} H\left(\tfrac{1}{4}, \tfrac{1}{2}, \tfrac{1}{8}, \tfrac{1}{8}\right) + \tfrac{1}{4} H\left(\tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}, \tfrac{1}{4}\right) + \tfrac{1}{4} H(1,0,0,0) \\ &= \tfrac{1}{4} \times \tfrac{7}{4} + \tfrac{1}{4} \times \tfrac{7}{4} + \tfrac{1}{4} \times 2 + \tfrac{1}{4} \times 0 \;=\; \frac{11}{8} \text{ bits.} \end{aligned}$$

By the same method, $H(Y|X) = \frac{13}{8}$ bits, and by the chain rule $H(X,Y) = H(X) + H(Y|X) = \frac{7}{4} + \frac{13}{8} = \frac{27}{8}$ bits (which also equals $H(Y) + H(X|Y) = 2 + \frac{11}{8} = \frac{27}{8}$, confirming the chain rule both ways). Note $H(Y|X) \neq H(X|Y)$ ($\frac{13}{8} \neq \frac{11}{8}$), as promised above.

An alternative worked example

The book's table is an excellent teaching device because every entry is a negative power of 2, so every entropy below comes out as an exact fraction with no calculator needed. Here is a second table with the same property, a different story, and a different (and, as we will see, more extreme) dependence structure, worked in full.

Example 2.7 (Alarm level and dashboard colour)**.** A monitoring system classifies each incoming event into one of four *alarm levels* $X \in \{1,2,3,4\}$ (from routine to critical) with

$$p(x) = \left(\tfrac{1}{2}, \tfrac{1}{4}, \tfrac{1}{8}, \tfrac{1}{8}\right),$$

the same marginal as theorem 2.6, chosen on purpose so that $H(X)$ is already known and the two examples are easy to compare. A simplified dashboard displays only a *colour* $Y \in \{\text{green}, \text{amber}, \text{red}\} = \{1,2,3\}$ summarising the alarm. Suppose the joint law is

$X\backslash Y$	green (1)	amber (2)	red (3)	$p(x)$
1	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{2}$
2	$\frac{1}{4}$	0	0	$\frac{1}{4}$
3	0	$\frac{1}{8}$	0	$\frac{1}{8}$
4	0	0	$\frac{1}{8}$	$\frac{1}{8}$
$p(y)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{4}$	1

Every entry is again a power of 2 (or 0), and the row sums recover the stated marginal of X. In words: routine events ($X = 1$) can show up as any colour, in the same proportions as the dashboard's overall colour frequencies, but every *elevated* alarm level ($X = 2, 3, 4$) maps to exactly one colour, so seeing $X \in \{2, 3, 4\}$ is fully diagnostic of Y.

Step 1: check it is a valid joint distribution. The twelve entries are nonnegative and sum to $\frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1$ (six nonzero cells, with the other six exact zeros, allowed by the usual convention $0 \log 0 = 0$).

Step 2: the marginals. Summing rows gives $p(x) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right)$ by construction, and summing columns gives $p(y) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$.

Step 3: $H(X)$ and $H(Y)$. These are read off the marginals exactly as in theorem 1.5:

$$H(X) = \tfrac{1}{2}(1) + \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{8}(3) = \tfrac{7}{4} \text{ bits}, \qquad H(Y) = \tfrac{1}{2}(1) + \tfrac{1}{4}(2) + \tfrac{1}{4}(2) = \tfrac{3}{2} \text{ bits}.$$

Step 4: $H(X, Y)$. There are six nonzero cells, with value $\frac{1}{4}$ at two of them and $\frac{1}{8}$ at the other four. Reading directly from the table: $(1,1) = \frac{1}{4}$, $(1,2) = \frac{1}{8}$, $(1,3) = \frac{1}{8}$, $(2,1) = \frac{1}{4}$, $(3,2) = \frac{1}{8}$, $(4,3) = \frac{1}{8}$. Using $-\log_2$ of each cell value ($-\log_2 \frac{1}{4} = 2$, $-\log_2 \frac{1}{8} = 3$):

$$\begin{aligned} H(X, Y) &= \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{8}(3) + \tfrac{1}{4}(2) + \tfrac{1}{8}(3) + \tfrac{1}{8}(3) \\ &= \tfrac{1}{2} + \tfrac{3}{8} + \tfrac{3}{8} + \tfrac{1}{2} + \tfrac{3}{8} + \tfrac{3}{8} \;=\; 1 + \tfrac{12}{8} \;=\; \tfrac{5}{2} \text{ bits}. \end{aligned}$$

Step 5: $H(Y|X)$, two ways. Directly: given $X = 1$ (probability $\frac{1}{2}$), Y has law $\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$ (normalising the first row by $\frac{1}{2}$), the same shape as the marginal of Y, so $H(Y \mid X = 1) = H(Y) = \frac{3}{2}$ bits. Given $X = 2, 3$, or 4 (probabilities $\frac{1}{4}, \frac{1}{8}, \frac{1}{8}$), Y is deterministic, so $H(Y \mid X = x) = 0$ for $x \in \{2, 3, 4\}$. Averaging,

$$H(Y|X) = \tfrac{1}{2}\left(\tfrac{3}{2}\right) + \tfrac{1}{4}(0) + \tfrac{1}{8}(0) + \tfrac{1}{8}(0) = \tfrac{3}{4} \text{ bits}.$$

Via the chain rule, $H(Y|X) = H(X, Y) - H(X) = \frac{5}{2} - \frac{7}{4} = \frac{3}{4}$ bits: the two routes agree.

Step 6: $H(X|Y)$, two ways. Directly: given $Y = 1$ (probability $\frac{1}{2}$), the two contributing cells are $(X, Y) = (1, 1) = \frac{1}{4}$ and $(2, 1) = \frac{1}{4}$, so $p(X \mid Y = 1) = \left(\frac{1}{2}, \frac{1}{2}, 0, 0\right)$ and $H(X \mid Y = 1) = 1$ bits. Given $Y = 2$ (probability $\frac{1}{4}$), the cells are $(1, 2) = \frac{1}{8}$ and $(3, 2) = \frac{1}{8}$, giving $p(X \mid Y = 2) = \left(\frac{1}{2}, 0, \frac{1}{2}, 0\right)$ and $H(X \mid Y = 2) = 1$ bits. Given $Y = 3$ (probability $\frac{1}{4}$), similarly $p(X \mid Y = 3) = \left(\frac{1}{2}, 0, 0, \frac{1}{2}\right)$ and $H(X \mid Y = 3) = 1$ bits. Every conditional law of X given Y is a two-point uniform distribution, so

$$H(X|Y) = \tfrac{1}{2}(1) + \tfrac{1}{4}(1) + \tfrac{1}{4}(1) = 1 \text{ bits}.$$

Via the chain rule, $H(X|Y) = H(X, Y) - H(Y) = \frac{5}{2} - \frac{3}{2} = 1$ bits: again the two routes agree.

Step 7: read off the asymmetry. $H(Y|X) = \frac{3}{4}$ bits $\neq 1$ bits $= H(X|Y)$: knowing the alarm level X leaves less residual uncertainty about the colour Y than knowing the colour leaves about the exact alarm level, which matches the story (routine events, $X = 1$, are common but colour-ambiguous, while the colour, once seen, still leaves a two-way ambiguity among the alarm levels that could have produced it).

Key idea: reading the table without computing anything

Two features of theorem 2.7 are visible directly from the table, and are worth more than the arithmetic. First, row $X = 1$ is exactly $\frac{1}{2}\, p(y)$: conditioning on $X = 1$ leaves Y distributed exactly as its unconditional law, so $X = 1$ is, in a precise sense, uninformative

about Y beyond what one already knew. Second, rows $X = 2, 3, 4$ each have a single nonzero entry: those alarm levels determine the colour outright, so all of the residual uncertainty $H(Y|X) = \frac{3}{4}$ bits comes entirely from the $X = 1$ branch, weighted by $p(X{=}1) = \frac{1}{2}$. Being able to read qualitative conclusions off the zero pattern of a table, before computing a single logarithm, is a skill worth having for the rest of the chapter.

2.1 Chain rule for several variables

For $X^n = (X_1, \dots, X_n)$, write $X^{i-1} = (X_1, \dots, X_{i-1})$, with X^0 empty.

Theorem 2.8 (Chain rule for entropy)**.**

$$H(X^n) = \sum_{i=1}^{n} H\big(X_i \big| X^{i-1}\big)\,.$$

Proof. Repeated application of theorem 2.3 gives $H(X_1, X_2) = H(X_1) + H(X_2|X_1)$, then $H(X_1, X_2, X_3) = H(X_1) + H(X_2|X_1) + H(X_3|X_2, X_1)$, and so on. Equivalently, factor the joint mass function as $p(x^n) = \prod_{i=1}^{n} p(x_i \mid x^{i-1})$, take $-\log$ of both sides, and average under p. The sum over $x_{i+1}, \dots, x_n$ of the i-th term marginalises it down to a function of x^i alone, leaving exactly $H\big(X_i \big| X^{i-1}\big)$. ∎

3 Relative entropy and mutual information

Definition 3.1 (Relative entropy, or Kullback–Leibler divergence)**.** [Kullback and Leibler, 1951]. Let p and q be mass functions on the same alphabet $\mathcal{X}$. The relative entropy of p with respect to q is

$$D(p\|q) = \sum_{x} p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_p\left[\log \frac{p(X)}{q(X)}\right],$$

with the zero-conventions of the Conventions above. In particular, $D(p\|q) = +\infty$ as soon as some x has $p(x) > 0 = q(x)$.

The notation is directional: p is the true, or averaging, law, while q is a candidate model. If we knew p exactly we could code its outcomes using, on average, $H(p)$ bits. If instead we code using a scheme optimised for q, the average cost rises to

$$\mathbb{E}_p[-\log q(X)] = H(p) + D(p\|q)$$

(Cover and Thomas, 2006, §2.3. This identity is proved directly at the end of the chain rule, theorem 4.4, and used again in section 10.) So $D(p\|q)$ is exactly the average number of *extra* bits paid for using q instead of the truth p.

Remark 3.2. Relative entropy is **not a metric**. In general $D(p\|q) \neq D(q\|p)$, and the triangle inequality fails. It is nonnegative (proved in theorem 5.3), which is enough to justify calling it a "distance" or "divergence" informally, but not enough to justify triangle-inequality-style reasoning with it.

Example 3.3 (Cover and Thomas, 2006, Example 2.3.1)**.** Let $\mathcal{X} = \{0, 1\}$, $p(1) = r$, $p(0) = 1 - r$, and $q(1) = s$, $q(0) = 1 - s$. Then

$$D(p\|q) = (1-r)\log\frac{1-r}{1-s} + r\log\frac{r}{s}, \qquad D(q\|p) = (1-s)\log\frac{1-s}{1-r} + s\log\frac{s}{r}.$$

If $r = s$ both vanish. If $r = \frac{1}{2}$, $s = \frac{1}{4}$,

$$D(p\|q) = \tfrac{1}{2}\log\tfrac{1/2}{3/4} + \tfrac{1}{2}\log\tfrac{1/2}{1/4} = 1 - \tfrac{1}{2}\log 3 \approx 0.2075\,\text{bits},$$

$$D(q\|p) = \tfrac{3}{4}\log\tfrac{3/4}{1/2} + \tfrac{1}{4}\log\tfrac{1/4}{1/2} = \tfrac{3}{4}\log 3 - 1 \approx 0.1887\,\text{bits},$$

so $D(p\|q) \neq D(q\|p)$, confirming the caution above with an explicit pair of numbers.

Exercise 3.4 (solves Cover and Thomas, 2006, Problem 2.35)**.** Let X take three values $\{a, b, c\}$, with $p = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$ and $q = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$ (uniform). Find $D(p\|q)$ and $D(q\|p)$.

Answer. $D(p\|q) = \frac{1}{2}\log\frac{3}{2} + \frac{1}{2}\log\frac{3}{4} = \log_2 3 - \frac{3}{2} \approx 0.0850$ bits, and $D(q\|p) = \frac{1}{3}\log\frac{2}{3} + \frac{2}{3}\log\frac{4}{3} = \frac{5}{3} - \log_2 3 \approx 0.0817$ bits. Close in value here, but still unequal, and computed against a *uniform* reference q this time, which is the comparison used throughout section 5 to bound entropy from above.

Definition 3.5 (Mutual information)**.** For jointly distributed X, Y,

$$\begin{aligned} I(X;Y) &= D(p_{XY}\|p_X p_Y) \\ &= \sum_{x,y} p_{XY}(x,y)\log\frac{p_{XY}(x,y)}{p_X(x)p_Y(y)}. \end{aligned}$$

Mutual information compares the actual joint law with the joint law that would hold under independence: it is a divergence, specialised to measuring *how far* (X, Y) *is from independence.*

Theorem 3.6 (Equivalent forms of mutual information)**.**

$$\begin{aligned} I(X;Y) &= H(X) - H(X|Y) \\ &= H(Y) - H(Y|X) \\ &= H(X) + H(Y) - H(X,Y) \\ &= I(Y;X)\,. \end{aligned}$$

In particular, $I(X;X) = H(X)$ *(entropy is sometimes called* self-information *for this reason).*

Proof. Using $p_{XY}(x,y) = p_{X|Y}(x \mid y)p_Y(y)$,

$$\begin{aligned} I(X;Y) &= \sum_{x,y} p_{XY}(x,y)\log\frac{p_{X|Y}(x \mid y)}{p_X(x)} \\ &= -\sum_x p_X(x)\log p_X(x) + \sum_{x,y} p_{XY}(x,y)\log p_{X|Y}(x \mid y) \\ &= H(X) - H(X|Y)\,. \end{aligned}$$

Symmetry, $I(X;Y) = I(Y;X)$, is immediate from theorem 3.5, since the defining ratio is symmetric in $x \leftrightarrow y$ once both p_{XY} and $p_X p_Y$ are. The joint-entropy identity then follows from the two-variable chain rule (theorem 2.3), and setting $Y = X$ gives $H(X|X) = 0$, so $I(X;X) = H(X) - 0 = H(X)$. ∎

Remark 3.7 (Revisiting the two worked examples). For theorem 2.6, $I(X;Y) = H(X) - H(X|Y) = \frac{7}{4} - \frac{11}{8} = \frac{3}{8}$ bits (Cover and Thomas, 2006, Example 2.4.1 states this value directly). For theorem 2.7, $I(X;Y) = H(X) - H(X|Y) = \frac{7}{4} - 1 = \frac{3}{4}$ bits, exactly twice as large, in agreement with the much stronger dependence built into that table (three of the four alarm levels there determine the colour outright, whereas in the book's table only one value of Y is fully diagnostic of X).

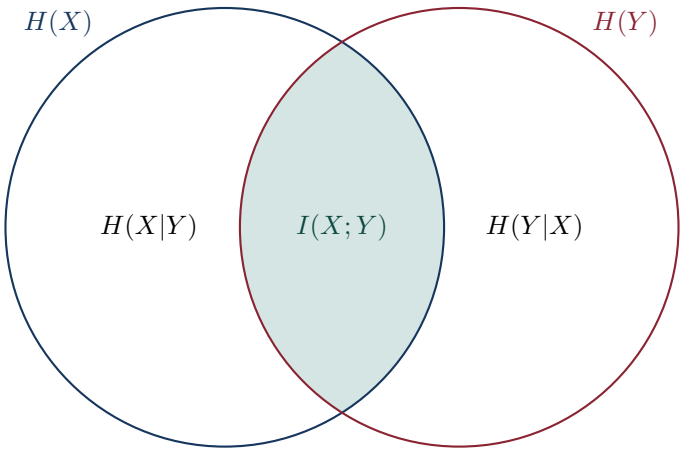


Figure 2: Entropy, conditional entropy, and mutual information, after Cover and Thomas [2006, Fig. 2.2]. The union of the two discs represents $H(X,Y)$, and the overlap represents $I(X;Y) = H(X)+H(Y)-H(X,Y)$. This picture is a genuine theorem for *two* random variables (the underlying signed measure of Yeung, 1991 is non-negative in this case) but, as **??** in section 7 shows, it stops being reliable once a third variable is added.

Example 3.8 (A binary symmetric channel)**.** Let $X \sim \text{Bernoulli}(1/2)$, let $N \sim \text{Bernoulli}(\varepsilon)$ be independent of X, and observe $Y = X \oplus N$ (a bit sent through a channel that flips it with probability ε). Then Y is fair and

$$H(Y) = 1\,\text{bits}, \qquad H(Y|X) = H(\varepsilon)\,,$$

because, given $X = x$, $Y = x \oplus N$ has exactly the law of N shifted, still with entropy $H(\varepsilon)$. Consequently,

$$I(X;Y) = 1 - H(\varepsilon)\,.$$

At $\varepsilon = 0$ the bit is transmitted perfectly and $I(X;Y) = 1\,\text{bits}$. At $\varepsilon = 1/2$ the output is independent of the input (Y is a fair coin regardless of X) and $I(X;Y) = 0$. We revisit this channel, called the *binary symmetric channel* (BSC), in section 7 and section 9, where $1-H(\varepsilon)$ turns out to be its capacity.

4 Chain rules for entropy, relative entropy, and mutual information

Theorem 2.8 already gave the chain rule for the entropy of several random variables. This section extends the same idea, factor the joint object into a product (or sum) over one variable at a time, to conditional mutual information and to relative entropy.

Definition 4.1 (Conditional mutual information)**.**

$$I(X;Y|Z) = H(X|Z) - H(X|Y,Z)\,.$$

Equivalently,

$$I(X;Y|Z) = \sum_z p_Z(z) D\big(p_{XY|Z=z} \big\| p_{X|Z=z} p_{Y|Z=z}\big)\,,$$

the average, over Z, of the mutual information between X and Y under the conditional law given $Z = z$.

Theorem 4.2 (Chain rule for mutual information)**.**

$$I(X^n;Y)=\sum_{i=1}^{n} I\big(X_i;Y\big|X^{i-1}\big)\,.$$

A frequently used two-block form (the case $n=2$, relabelling $X_1=Z$, $X_2=Y$, $Y=X$, useful enough on its own to be worth naming) is

$$I(X;Y,Z)=I(X;Z)+I(X;Y|Z)\,.$$

Proof. Subtract the conditional entropy chain rule from the unconditional one:

$$\begin{aligned} I(X^n;Y) &= H(X^n)-H(X^n|Y)\\ &= \sum_{i=1}^{n} H\big(X_i\big|X^{i-1}\big)-\sum_{i=1}^{n} H\big(X_i\big|X^{i-1},Y\big)\\ &= \sum_{i=1}^{n}\big[H\big(X_i\big|X^{i-1}\big)-H\big(X_i\big|X^{i-1},Y\big)\big]=\sum_{i=1}^{n} I\big(X_i;Y\big|X^{i-1}\big)\,. \end{aligned}$$

■

Definition 4.3 (Conditional relative entropy)**.** Given conditional mass functions $p_{Y|X}$, $q_{Y|X}$ and an averaging law p_X, define

$$D(p_{Y|X}\|q_{Y|X}\mid p_X)=\sum_x p_X(x)D\big(p_{Y|X=x}\big\|q_{Y|X=x}\big)\,.$$

Theorem 4.4 (Chain rule for relative entropy)**.** *Let $p_{XY}=p_Xp_{Y|X}$ and $q_{XY}=q_Xq_{Y|X}$. Then*

$$D(p_{XY}\|q_{XY})=D(p_X\|q_X)+D(p_{Y|X}\|q_{Y|X}\mid p_X).$$

Proof. On the support of p_{XY},

$$\log\frac{p_{XY}(x,y)}{q_{XY}(x,y)}=\log\frac{p_X(x)}{q_X(x)}+\log\frac{p_{Y|X}(y\mid x)}{q_{Y|X}(y\mid x)}.$$

Average both sides under p_{XY}. ■

Remark 4.5. Setting $q_X=p_X$ recovers the coding-theoretic identity from section 3: if $q_{XY}=p_Xq_{Y|X}$ agrees with p_{XY} on the marginal of X but uses a possibly wrong conditional model $q_{Y|X}$, then $D(p_{XY}\|q_{XY})=D(p_{Y|X}\|q_{Y|X}\mid p_X)\geq 0$ (nonnegativity is theorem 5.3 below), with equality exactly when the conditional model is correct almost surely. Section 10 uses the same chain rule again, in the special case $q=p$, to connect entropy with the probability that two independent draws collide.

5 Jensen's inequality and the information inequality

Definition 5.1 (Convexity)**.** A function f on an interval is convex if, for $0\leq\lambda\leq 1$,

$$f(\lambda x+(1-\lambda)y)\leq\lambda f(x)+(1-\lambda)f(y).$$

It is strictly convex when equality for $0<\lambda<1$ forces $x=y$. A function is concave if its negative is convex.

If f is twice differentiable, $f'' \geq 0$ on an interval implies convexity there, and $f'' > 0$ implies strict convexity (Cover and Thomas, 2006, Thm. 2.6.1, by Taylor's theorem with remainder). Two examples used throughout this chapter:

$$-\log t \text{ is strictly convex on } (0,\infty), \qquad -t\log t \text{ is strictly concave on } (0,\infty).$$

Theorem 5.2 (Jensen's inequality)**.** *[Cover and Thomas, 2006, Theorem 2.6.2], tracing back to Jensen [1906] in far greater generality than the finite-alphabet case proved below. If f is convex and W is a finite-valued random variable, then*

$$f(\mathbb{E}W) \leq \mathbb{E}f(W).$$

If f is strictly convex, equality holds if and only if W is constant almost surely.

Proof. For W taking values $w_1, \ldots, w_m$ with probabilities $\lambda_1, \ldots, \lambda_m$, repeated application of the two-point definition of convexity gives

$$f\left(\sum_i \lambda_i w_i\right) \leq \sum_i \lambda_i f(w_i).$$

For strictly convex f, equality forces every positive-probability w_i to coincide (otherwise some two-point step in the induction is strict). ■

Theorem 5.3 (Information inequality, or Gibbs' inequality)**.** *[Cover and Thomas, 2006, Theorem 2.6.3]. For mass functions p, q on the same alphabet,*

$$D(p\|q) \geq 0,$$

with equality if and only if $p = q$.

Proof. If p is not absolutely continuous with respect to q, $D(p\|q) = +\infty$ and there is nothing to prove. Otherwise, apply Jensen's inequality to the strictly concave function log, under p, on the support $A = \{x : p(x) > 0\}$ of p:

$$\begin{aligned} -D(p\|q) &= \sum_{x\in A} p(x)\log\frac{q(x)}{p(x)} \\ &\leq \log\sum_{x\in A} q(x) \leq \log\sum_{x\in\mathcal{X}} q(x) = \log 1 = 0. \end{aligned}$$

For equality, both inequalities must be equalities. Strict concavity of log forces $q(x)/p(x)$ to be constant, say c, on A, so $\sum_{x\in A} q(x) = c$. Equality in the second step forces q to put no mass outside A, so $c = \sum_{x\in A} q(x) = \sum_{x\in\mathcal{X}} q(x) = 1$. Hence $q(x) = p(x)$ on A, and q vanishes off A exactly where p does, so $p = q$ everywhere. ■

Remark 5.4 (Why "Gibbs' inequality"?). Gibbs worked with the continuous entropy functional of classical statistical mechanics, not with a sum over a finite alphabet, and did not himself state theorem 5.3 in this discrete, probabilistic form. The connection is that his variational characterisation of equilibrium, that at fixed average energy the Boltzmann distribution uniquely minimises what is now called the free energy, is, once translated into the language of probability vectors, exactly the statement that relative entropy between any competing distribution and the Boltzmann distribution cannot be negative. The name records this common variational ancestry rather than a claim that Gibbs proved the inequality in the form stated here.

Remark 5.5 (Pinsker's inequality: a quantitative strengthening). Theorem 5.3 says $D(p\|q) = 0$ forces $p = q$, but says nothing about *how close* p and q must be when $D(p\|q)$ is merely small. Pinsker's inequality answers this. Writing the total variation distance as $\|p - q\|_{TV} = \frac{1}{2}\sum_x |p(x) - q(x)|$ and taking logarithms to base e just for this statement (as is conventional here),

$$\|p - q\|_{TV} \;\le\; \sqrt{\tfrac{1}{2} D(p\|q)}\,.$$

So convergence in relative entropy is a strictly stronger requirement than convergence in total variation, a fact used constantly in information-theoretic proofs of concentration and convergence results beyond this chapter. The inequality is usually named after Pinsker [1964] (the English translation of a 1960 Russian original), though bounds of the same flavour, relating $D(p\|q)$ to a distance between p and q, were found independently around the same time by Kullback [1967], and the inequality also fits inside the broader theory of f-divergences developed by Csiszár [1967]. We do not prove it here: the shortest proofs either specialise first to a binary alphabet and use calculus, or go via the data-processing inequality for f-divergences (section 7), and either route would add machinery this chapter does not otherwise need.

Corollary 5.6 (Nonnegativity of mutual information)**.**

$$I(X;Y) \ge 0,$$

with equality if and only if $X \perp\!\!\!\perp Y$*. More generally,*

$$I(X;Y|Z) \ge 0,$$

with equality if and only if $X \perp\!\!\!\perp Y \mid Z$*.*

Proof. Apply theorem 5.3 to $I(X;Y) = D(p_{XY}\|p_X p_Y)$, which vanishes iff $p_{XY} = p_X p_Y$, i.e. iff $X \perp\!\!\!\perp Y$. For the conditional statement, apply it instead to each conditional law $p_{XY|Z=z}$ against $p_{X|Z=z}p_{Y|Z=z}$. ∎

Theorem 5.7 (Maximum entropy on a finite alphabet)**.** *[Cover and Thomas, 2006, Theorem 2.6.4]. If* X *takes values in an alphabet* $\mathcal{X}$ *of size* m*, then*

$$H(X) \le \log m,$$

with equality if and only if X *is uniform on* $\mathcal{X}$*.*

Proof. Let $u(x) = 1/m$ be the uniform law on $\mathcal{X}$. Then

$$D(p_X\|u) = \sum_x p_X(x)\log(m p_X(x)) = \log m - H(X) \;\ge\; 0$$

by theorem 5.3, with equality iff $p_X = u$. ∎

Example 5.8 (A loaded die)**.** For a fair six-sided die, $H(X) = \log_2 6 \approx 2.585$ bits, attaining the bound of theorem 5.7 with equality. For a die loaded so that $p = \left(\frac{1}{2}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}, \frac{1}{10}\right)$,

$$H(X) = -\tfrac{1}{2}\log_2\tfrac{1}{2} - 5 \times \tfrac{1}{10}\log_2\tfrac{1}{10} = \tfrac{1}{2} + \tfrac{1}{2}\log_2 10 \approx 2.161 \text{ bits},$$

strictly below $\log_2 6 \approx 2.585$ bits, as theorem 5.7 requires: any departure from uniformity can only lower the entropy, never raise it.

Theorem 5.9 (Conditioning reduces entropy, on average)**.** *[Cover and Thomas, 2006, Theorem 2.6.5].*

$$H(X|Y) \le H(X),$$

with equality if and only if $X \perp\!\!\!\perp Y$.

Proof. By theorem 3.6, $H(X) - H(X|Y) = I(X;Y)$, which is ≥ 0 by theorem 5.6, with equality iff $X \perp\!\!\!\perp Y$. ■

Remark 5.10. The inequality is a statement about an *average*. A particular observation $Y = y$ can increase uncertainty about X: $H(X \mid Y = y) > H(X)$ is possible for some y, even though $H(X|Y) = \sum_y p(y) H(X \mid Y = y) \le H(X)$ always.

Example 5.11 (Cover and Thomas, 2006, Example 2.6.1)**.** Let (X, Y) have joint law $p(1,1) = 0$, $p(2,1) = \frac{3}{4}$, $p(1,2) = \frac{1}{8}$, $p(2,2) = \frac{1}{8}$ (so $p(x) = \left(\frac{1}{8}, \frac{7}{8}\right)$ and $p(y) = \left(\frac{3}{4}, \frac{1}{4}\right)$). Then

$$H(X) = H\left(\tfrac{1}{8}\right) \approx 0.544 \,\text{bits}, \qquad H(X \mid Y = 1) = 0 \,\text{bits}, \qquad H(X \mid Y = 2) = 1 \,\text{bits},$$

since $Y = 1$ pins X down to $X = 2$ exactly, while $Y = 2$ leaves X uniform on $\{1, 2\}$. Averaging,

$$H(X|Y) = \tfrac{3}{4}(0) + \tfrac{1}{4}(1) = \tfrac{1}{4} \,\text{bits}.$$

So observing $Y = 1$ *decreases* the uncertainty in X all the way to zero, while observing $Y = 2$ *increases* it, from 0.544 to 1 bits, exactly the phenomenon of theorem 5.10. Yet the average $\frac{1}{4}$ bits is still below $H(X) \approx 0.544$ bits, as theorem 5.9 guarantees.

Corollary 5.12 (Independence bound on entropy)**.**

$$H(X_1, \dots, X_n) \le \sum_{i=1}^{n} H(X_i),$$

with equality if and only if $X_1, \dots, X_n$ *are mutually independent.*

Proof. Combine the entropy chain rule (theorem 2.8) with $H\left(X_i \middle| X^{i-1}\right) \le H(X_i)$ (theorem 5.9, applied with $Y = X^{i-1}$). Equality throughout is equivalent to $X_i \perp\!\!\!\perp X^{i-1}$ for every i, which is equivalent to mutual independence of $X_1, \dots, X_n$. ■

Remark 5.13. The equality condition needs *mutual* independence, not merely pairwise independence. The classical counterexample is X, Y independent fair bits and $Z = X \oplus Y$: any two of $\{X, Y, Z\}$ are independent, but $H(X, Y, Z) = 2 < 3 = H(X) + H(Y) + H(Z)$, so the bound of theorem 5.12 is strict even though every *pair* is independent.

6 The log-sum inequality and convexity

Theorem 6.1 (Log-sum inequality)**.** *[Cover and Thomas, 2006, Theorem 2.7.1]. For nonnegative numbers* $a_1, \dots, a_n$ *and* $b_1, \dots, b_n$,

$$\sum_{i=1}^{n} a_i \log \frac{a_i}{b_i} \;\ge\; \left(\sum_{i=1}^{n} a_i\right) \log \frac{\sum_{i=1}^{n} a_i}{\sum_{i=1}^{n} b_i},$$

with the usual conventions $0 \log 0 = 0$, $0 \log \frac{0}{b} = 0$, *and* $a \log \frac{a}{0} = +\infty$ *for* $a > 0$. *Equality holds if and only if* a_i / b_i *is the same for every* i *with* $b_i > 0$.

Proof. Assume first $a_i, b_i > 0$ for every i. Let $B = \sum_i b_i$, put $\alpha_i = b_i/B$ (so $\alpha_i \geq 0$, $\sum_i \alpha_i = 1$), and put $t_i = a_i/b_i$. Since $f(t) = t \log t$ is strictly convex on $(0, \infty)$ (theorem 5.2 above, or directly $f''(t) = \frac{1}{t} \log e > 0$), Jensen's inequality applied to the random variable taking value t_i with probability α_i gives

$$\sum_i \alpha_i f(t_i) \;\geq\; f\left(\sum_i \alpha_i t_i\right).$$

The left side equals $\sum_i \frac{b_i}{B} \cdot \frac{a_i}{b_i} \log \frac{a_i}{b_i} = \frac{1}{B} \sum_i a_i \log \frac{a_i}{b_i}$. The right side equals $f\big(\frac{1}{B} \sum_i a_i\big) = \frac{A}{B} \log \frac{A}{B}$, where $A = \sum_i a_i$. Multiplying through by B gives the claim. Strict convexity of f makes the Jensen step an equality exactly when every t_i with $\alpha_i > 0$ coincides, that is, exactly when a_i/b_i is constant over the indices with $b_i > 0$. The boundary cases ($a_i = 0$ or $b_i = 0$ for some i) follow by continuity, using the stated conventions. ∎

Example 6.2 (Two small numerical illustrations)**.** Take $n = 2$. With $a = (2, 6)$ and $b = (1, 1)$,

$$\sum_i a_i \log_2 \frac{a_i}{b_i} = 2 \log_2 2 + 6 \log_2 6 = 2 + 6 \log_2 6 \approx 17.510,$$

while the right side is

$$\left(\sum_i a_i\right) \log_2 \frac{\sum_i a_i}{\sum_i b_i} = 8 \log_2 \frac{8}{2} = 8 \log_2 4 = 16,$$

a strict inequality, $17.510 > 16$, since $a_1/b_1 = 2 \neq 6 = a_2/b_2$. Now take $a = (2, 4)$ and $b = (1, 2)$, so that the ratios $a_i/b_i = 2$ agree for both i. The left side is $2 \log_2 2 + 4 \log_2 2 = 2 + 4 = 6$, and the right side is $6 \log_2 \frac{6}{3} = 6 \log_2 2 = 6$: equality, exactly as theorem 6.1 predicts when the ratios are constant.

Remark 6.3 (A second proof that relative entropy is nonnegative). Theorem 5.3 was proved directly from Jensen's inequality applied to log. The log-sum inequality gives a second, purely algebraic route to the same fact, essentially by specialising theorem 6.1 to $a_x = p(x)$, $b_x = q(x)$:

$$D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \;\geq\; \Big(\textstyle\sum_x p(x)\Big) \log \frac{\sum_x p(x)}{\sum_x q(x)} = 1 \cdot \log \tfrac{1}{1} = 0,$$

with equality iff $p(x)/q(x)$ is constant on the support of p, which, exactly as in the proof of theorem 5.3, forces the constant to be 1 and hence $p = q$. Cover and Thomas present the material in the opposite order from the present one: the log-sum inequality first (Cover and Thomas, 2006, §2.7), then this computation as a reproof of nonnegativity. We instead proved nonnegativity directly from Jensen's inequality in section 5, because it was needed there already for theorem 5.6 and theorem 5.7, before the log-sum inequality had been introduced. Either proof is complete on its own. Seeing both is what makes the log-sum inequality feel like more than a technical lemma invented for the next two theorems.

Theorem 6.4 (Joint convexity of relative entropy)**.** *[Cover and Thomas, 2006, Theorem 2.7.2]. For probability mass functions p_1, p_2, q_1, q_2 on the same alphabet and $0 \leq \lambda \leq 1$,*

$$D(\lambda p_1 + (1-\lambda)p_2 \| \lambda q_1 + (1-\lambda)q_2) \;\leq\; \lambda D(p_1\|q_1) + (1-\lambda) D(p_2\|q_2)\,.$$

Proof. Fix x and apply theorem 6.1 with $n = 2$, $a_1 = \lambda p_1(x)$, $a_2 = (1-\lambda)p_2(x)$, $b_1 = \lambda q_1(x)$, $b_2 = (1-\lambda)q_2(x)$:

$$\big(\lambda p_1(x) + (1-\lambda)p_2(x)\big) \log \frac{\lambda p_1(x) + (1-\lambda)p_2(x)}{\lambda q_1(x) + (1-\lambda)q_2(x)}$$

$$\leq \lambda p_1(x) \log \frac{p_1(x)}{q_1(x)} + (1-\lambda) p_2(x) \log \frac{p_2(x)}{q_2(x)}.$$

Summing over $x \in \mathcal{X}$ gives the claim. ■

Theorem 6.5 (Concavity of entropy)**.** *[Cover and Thomas, 2006, Theorem 2.7.3]. $H(p)$ is a concave function of p: for probability mass functions p_1, p_2 on a common alphabet of size m and $0 \leq \lambda \leq 1$,*

$$H(\lambda p_1 + (1-\lambda)p_2) \;\geq\; \lambda H(p_1) + (1-\lambda) H(p_2).$$

Proof. Write $H(p) = \log m - D(p\|u)$, where u is uniform on the common alphabet (this is exactly theorem 5.7's proof, rearranged). Since u does not depend on p, applying theorem 6.4 with $q_1 = q_2 = u$ gives

$$D(\lambda p_1 + (1-\lambda)p_2 \| u) \leq \lambda D(p_1\|u) + (1-\lambda) D(p_2\|u)\,,$$

and subtracting both sides from $\log m$ reverses the inequality and gives concavity of H. ■

Alternative proof. Let θ be a $\{1,2\}$-valued random variable, independent of everything else, with $\mathbb{P}(\theta = 1) = \lambda$, $\mathbb{P}(\theta = 2) = 1 - \lambda$. Let $X_1 \sim p_1$ and $X_2 \sim p_2$ be defined on the same alphabet, and set $Z = X_\theta$ (that is, $Z = X_1$ if $\theta = 1$ and $Z = X_2$ if $\theta = 2$). Then Z has law $\lambda p_1 + (1-\lambda)p_2$, and $H(Z \mid \theta = 1) = H(p_1)$, $H(Z \mid \theta = 2) = H(p_2)$, so

$$H(Z|\theta) = \lambda H(p_1) + (1-\lambda) H(p_2).$$

Theorem 5.9, "conditioning cannot increase entropy on average", gives $H(Z) \geq H(Z|\theta)$, which is exactly the claimed inequality. ■

Example 6.6 (Mixing a fair and a biased coin)**.** Let $p_1 = \left(\frac{1}{2}, \frac{1}{2}\right)$ (a fair coin) and $p_2 = \left(\frac{1}{10}, \frac{9}{10}\right)$ (a heavily biased coin), and mix them with $\lambda = \frac{1}{2}$. The mixture is $p_{\text{mix}} = \left(\frac{3}{10}, \frac{7}{10}\right)$, and

$$H(p_{\text{mix}}) = H\left(\tfrac{3}{10}\right) \approx 0.881 \,\text{bits}, \qquad \tfrac{1}{2}H(p_1) + \tfrac{1}{2}H(p_2) = \tfrac{1}{2}(1) + \tfrac{1}{2}H\left(\tfrac{1}{10}\right) \approx 0.734 \,\text{bits}.$$

As theorem 6.5 guarantees, $0.881 > 0.734$: not observing which coin was tossed leaves *more* uncertainty than the average uncertainty of the two coins separately, because not knowing θ is itself an additional source of randomness.

Theorem 6.7 (Concavity and convexity of mutual information)**.** *[Cover and Thomas, 2006, Theorem 2.7.4]. Let $(X, Y) \sim p(x, y) = p(x)p(y \mid x)$. Then $I(X;Y)$ is a concave function of $p(x)$ for fixed channel $p(y \mid x)$, and a convex function of the channel $p(y \mid x)$ for fixed $p(x)$.*

Proof. For the first part, expand

$$I(X;Y) = H(Y) - H(Y|X) = H(Y) - \sum_x p(x) H(Y \mid X = x).$$

With $p(y \mid x)$ held fixed, $p(y) = \sum_x p(x)p(y \mid x)$ is a linear function of $p(x)$, so $H(Y)$, a concave function of the law of Y (theorem 6.5), is a concave function of $p(x)$ by composition with a linear map. The second term is linear in $p(x)$, since each $H(Y \mid X = x)$ is a fixed number, weighted linearly. A concave function minus a linear function is concave, proving the first claim.

For the second part, fix $p(x)$ and let $p_1(y \mid x)$, $p_2(y \mid x)$ be two channels, with corresponding joints $p_i(x, y) = p(x)p_i(y \mid x)$ and Y-marginals $p_i(y)$, $i = 1, 2$. For $0 \leq \lambda \leq 1$ let $p_\lambda(y \mid x) = \lambda p_1(y \mid x) + (1-\lambda) p_2(y \mid x)$. Both the joint law and the Y-marginal mix in the same proportions,

$p_\lambda(x,y) = \lambda p_1(x,y) + (1-\lambda)p_2(x,y)$ and $p_\lambda(y) = \lambda p_1(y) + (1-\lambda)p_2(y)$, because $p(x)$ is common to all three channels. Writing $q_i(x,y) = p(x)p_i(y)$ for the corresponding independence coupling, $q_\lambda = \lambda q_1 + (1-\lambda)q_2$ as well. Since $I(X;Y) = D(p_{XY}\|p_X p_Y)$ (theorem 3.5), we have $I_\lambda(X;Y) = D(p_\lambda\|q_\lambda)$, and theorem 6.4 applied to this pair of mixtures gives

$$I_\lambda(X;Y) = D(p_\lambda\|q_\lambda) \le \lambda D(p_1\|q_1) + (1-\lambda)D(p_2\|q_2) = \lambda I_1(X;Y) + (1-\lambda)I_2(X;Y),$$

which is convexity of $I(X;Y)$ in the channel. ■

Key idea: why this pattern matters

Theorem 6.7 looks like a technical curiosity until it meets an optimisation problem. Channel capacity, defined in later chapters as $C = \max_{p(x)} I(X;Y)$ for a fixed channel, is then a maximisation of a *concave* function over the convex, compact simplex of input laws, so it has no spurious local maxima and standard tools apply. The convexity half is the corresponding fact for the channel design problem at fixed input. Both halves together are exactly the structure exploited by the Blahut–Arimoto algorithm for numerically computing channel capacity and rate–distortion functions [Blahut, 1972, Arimoto, 1972], an alternating-optimisation scheme that provably converges precisely because of this concave/convex split. None of this is needed to finish the present chapter.

7 Data processing

Definition 7.1 (Markov chain)**.** Random variables X, Y, Z form a Markov chain, written $X \to Y \to Z$, if the conditional law of Z given (X,Y) depends on Y alone, that is, if

$$p(x,y,z) = p(x)\,p(y \mid x)\,p(z \mid y).$$

Three consequences follow directly from theorem 7.1 and are used freely below.

- $X \to Y \to Z$ holds if and only if $X \perp\!\!\!\perp Z \mid Y$: Markovity implies conditional independence because
$$p(x,z \mid y) = \frac{p(x,y)p(z \mid y)}{p(y)} = p(x \mid y)\,p(z \mid y).$$
- $X \to Y \to Z$ implies $Z \to Y \to X$, since the defining factorisation is symmetric under reversing the chain (both directions reduce to $X \perp\!\!\!\perp Z \mid Y$). For this reason the condition is often written $X \leftrightarrow Y \leftrightarrow Z$.
- If $Z = f(Y)$ for a deterministic function f, then $X \to Y \to Z$ automatically, for any joint law of (X,Y).

Theorem 7.2 (Data-processing inequality)**.** *[Cover and Thomas, 2006, Theorem 2.8.1]. If $X \to Y \to Z$, then*

$$I(X;Y) \;\ge\; I(X;Z)\,.$$

Proof. By the chain rule for mutual information (theorem 4.2), the three-variable quantity $I(X;Y,Z)$ can be expanded in two orders:

$$I(X;Y,Z) = I(X;Z) + I(X;Y|Z) = I(X;Y) + I(X;Z|Y)\,.$$

Since $X \to Y \to Z$ gives $X \perp\!\!\!\perp Z \mid Y$, theorem 5.6 makes $I(X;Z|Y) = 0$. Since $I(X;Y|Z) \geq 0$ always (again theorem 5.6), equating the two expansions gives

$$I(X;Z) \;=\; I(X;Y) + I(X;Z|Y) - I(X;Y|Z) \;=\; I(X;Y) - I(X;Y|Z) \;\leq\; I(X;Y)\,.$$

Equality holds if and only if $I(X;Y|Z) = 0$, that is, if and only if $X \to Z \to Y$ is *also* a Markov chain. By the symmetric roles of X and Z in the hypothesis, the same argument gives $I(Y;Z) \geq I(X;Z)$. ■

Corollary 7.3. *If $Z = g(Y)$ for a deterministic (possibly random, applied independently of everything else) function g, then $I(X;Y) \geq I(X;g(Y))$.*

Proof. $X \to Y \to g(Y)$ is a Markov chain, by the third consequence of theorem 7.1 above, so theorem 7.2 applies. ■

Corollary 7.4. *If $X \to Y \to Z$, then $I(X;Y|Z) \leq I(X;Y)$.*

Proof. From the proof of theorem 7.2, $I(X;Y|Z) = I(X;Y) - I(X;Z)$, and $I(X;Z) \geq 0$ by theorem 5.6. ■

8 Sufficient statistics

This section is a short detour, following Cover and Thomas [2006, §2.9], showing what the data-processing inequality has to say about a central idea in statistics. Let $\{f_\theta(x)\}$ be a family of mass functions indexed by a parameter θ, let X be a sample from f_θ, and let $T = T(X)$ be any statistic, that is, any function of the sample, such as a sample mean or a sample variance. Since T is a deterministic function of X, $\theta \to X \to T(X)$ is a Markov chain (the third consequence of theorem 7.1), so theorem 7.3 gives

$$I(\theta;T(X)) \;\leq\; I(\theta;X)$$

for every distribution placed on θ. Some statistics lose no information at all.

Definition 8.1 (Sufficient statistic)**.** A statistic $T(X)$ is *sufficient* for the family $\{f_\theta(x)\}$ if $\theta \to T(X) \to X$ is a Markov chain for every distribution on θ, equivalently (theorem 7.1) if X is independent of θ given $T(X)$.

Proposition 8.2. *$T(X)$ is sufficient for $\{f_\theta(x)\}$ if and only if*

$$I(\theta;T(X)) = I(\theta;X)$$

for every distribution on θ, that is, if and only if equality holds in the data-processing inequality applied to $\theta \to X \to T(X)$.

Proof. Immediate from the equality condition in theorem 7.2, applied to the chain $\theta \to X \to T(X)$: equality holds there exactly when the reverse chain $\theta \to T(X) \to X$ also holds, which is theorem 8.1. ■

Example 8.3 (Sufficiency of the sample sum for a Bernoulli parameter)**.** Let $X_1, \ldots, X_n$ be i.i.d. Bernoulli(θ), and let $T(X_1, \ldots, X_n) = \sum_{i=1}^n X_i$. For a fixed number of successes k, every

sequence $(x_1, \dots, x_n) \in \{0,1\}^n$ with $\sum_i x_i = k$ has the same probability, $\theta^k(1-\theta)^{n-k}$, and there are $\binom{n}{k}$ such sequences, so

$$\mathbb{P}\big((X_1, \dots, X_n) = (x_1, \dots, x_n) \;\big|\; T = k\big) = \frac{\theta^k(1-\theta)^{n-k}}{\binom{n}{k}\,\theta^k(1-\theta)^{n-k}} = \frac{1}{\binom{n}{k}}$$

for every $(x_1, \dots, x_n)$ with $\sum_i x_i = k$, a number that does not depend on θ at all. *Given* the count $T = k$, every arrangement of the k successes among the n trials is equally likely, regardless of θ, which is exactly $\theta \perp\!\!\!\perp (X_1, \dots, X_n) \mid T$, so T is sufficient for θ (Cover and Thomas, 2006, §2.9). With $n = 3$, $k = 1$ concretely: each of the three sequences $(1,0,0)$, $(0,1,0)$, $(0,0,1)$ has conditional probability exactly $\frac{1}{3} = 1/\binom{3}{1}$, for every θ.

Two further families, treated fully in Cover and Thomas [2006, §2.9] and only stated here: for $X_1, \dots, X_n$ i.i.d. $\mathcal{N}(\theta, 1)$, the sample mean $\overline{X}_n$ is sufficient for θ, and for $X_1, \dots, X_n$ i.i.d. Uniform$(\theta, \theta+1)$, the pair $(\max_i X_i, \min_i X_i)$ is sufficient for θ. Both involve continuous random variables, for which entropy and mutual information are defined somewhat differently (Cover and Thomas, 2006, Ch. 8, differential entropy), which is why they are only stated and not proved here.

Definition 8.4 (Minimal sufficient statistic)**.** [Lehmann and Scheffé, 1950]. A sufficient statistic $T(X)$ is *minimal* if it is a function of every other sufficient statistic $U(X)$, that is, if

$$\theta \to T(X) \to U(X) \to X$$

for every sufficient U. A minimal sufficient statistic compresses the sample as much as possible without losing information about θ. Other sufficient statistics may carry extra, irrelevant randomness on top: for a normal mean, the pair consisting of the mean of the odd-indexed samples and the mean of the even-indexed samples is sufficient but not minimal, since $\overline{X}_n$ alone is already sufficient and is a function of that pair.

Theorem 8.5 (Factorisation criterion)**.** *Let $\{f_\theta(x) : \theta \in \Theta\}$ be a family of mass functions on a common finite or countable alphabet. A statistic $T(X)$ is sufficient for θ if and only if f_θ factors as*

$$f_\theta(x) = g\big(T(x), \theta\big)\, h(x)$$

for some functions $g \geq 0$ and $h \geq 0$, where h does not depend on θ.

Proof sketch, discrete case. If $T(X)$ is sufficient, $\mathbb{P}(X = x \mid T(X) = T(x))$ does not depend on θ. Call it $h(x)$, and set $f_\theta(x) = \mathbb{P}_\theta(T(X) = T(x)) \cdot h(x)$, which has the stated form with $g(t, \theta) = \mathbb{P}_\theta(T(X) = t)$. Conversely, if f_θ factors as stated, summing over $\{x' : T(x') = t\}$ gives $\mathbb{P}_\theta(T(X) = t) = g(t,\theta) \sum_{x' : T(x') = t} h(x')$, and dividing shows that

$$\mathbb{P}_\theta(X = x \mid T(X) = t) = \frac{h(x)}{\sum_{x' : T(x') = t} h(x')}$$

does not depend on θ, which is sufficiency. ∎

9 Fano's inequality

Suppose we observe Y and wish to guess a correlated random variable X with alphabet $\mathcal{X}$. From Y we compute an estimate $\widehat{X} = g(Y)$, where g may itself be randomised and $\widehat{X}$ may

take values in an alphabet $\widehat{\mathcal{X}}$ different from $\mathcal{X}$. Then $X \to Y \to \widehat{X}$ is a Markov chain. Write the probability of error as

$$P_e = \mathbb{P}(\widehat{X} \neq X).$$

If $H(X|Y) = 0$, then X is (almost surely) a function of Y (Cover and Thomas, 2006, Problem 2.5), so it can be guessed with $P_e = 0$. Fano's inequality [Fano, 1961] is the quantitative version of this fact: it bounds P_e from below whenever $H(X|Y)$ is bounded away from 0, and is one of the main tools used, in later chapters, to prove that certain communication rates are *not* achievable (Cover and Thomas, 2006, §2.10. It drives the converse to the channel coding theorem in Cover and Thomas's Chapter 7.)

Theorem 9.1 (Fano's inequality)**.** *[Cover and Thomas, 2006, Theorem 2.10.1]. For any estimator $\widehat{X} = g(Y)$ with $X \to Y \to \widehat{X}$ and $P_e = \mathbb{P}(\widehat{X} \neq X)$,*

$$H(P_e) + P_e \log |\mathcal{X}| \;\geq\; H(X \mid \widehat{X}) \;\geq\; H(X|Y)\,.$$

Proof. Define the error indicator $E = \mathbf{1}\{\widehat{X} \neq X\}$, so $H(E) = H(P_e)$. Expand $H(E, X \mid \widehat{X})$ by the conditional entropy chain rule in two orders:

$$\begin{aligned} H(E, X \mid \widehat{X}) &= H(X \mid \widehat{X}) + H(E \mid X, \widehat{X}) \\ &= H(E \mid \widehat{X}) + H(X \mid E, \widehat{X}). \end{aligned}$$

Since E is a deterministic function of $(X, \widehat{X})$, $H(E \mid X, \widehat{X}) = 0$. Since conditioning reduces entropy (theorem 5.9), $H(E \mid \widehat{X}) \leq H(E) = H(P_e)$. For the remaining term, condition on the value of E. Given $E = 0$, $\widehat{X} = X$ exactly, so $H(X \mid \widehat{X}, E{=}0) = 0$. Given $E = 1$, X ranges over at most $|\mathcal{X}|$ values, so $H(X \mid \widehat{X}, E{=}1) \leq \log |\mathcal{X}|$. Averaging,

$$H(X \mid E, \widehat{X}) = (1 - P_e) \cdot 0 + P_e \cdot H(X \mid \widehat{X}, E{=}1) \leq P_e \log |\mathcal{X}|.$$

Combining, $H(X \mid \widehat{X}) = H(E, X \mid \widehat{X}) - H(E \mid X, \widehat{X}) \leq H(E \mid \widehat{X}) + H(X \mid E, \widehat{X}) \leq H(P_e) + P_e \log |\mathcal{X}|$, which is the first inequality. For the second, theorem 7.3 applied to $X \to Y \to \widehat{X}$ gives $I\left(X; \widehat{X}\right) \leq I(X;Y)$, that is, $H(X) - H(X \mid \widehat{X}) \leq H(X) - H(X|Y)$, so $H(X \mid \widehat{X}) \geq H(X|Y)$. ∎

Remark 9.2. Setting $P_e = 0$ in theorem 9.1 forces $H(X|Y) \leq H(X \mid \widehat{X}) \leq H(0) + 0 = 0$, recovering the fact used to motivate the theorem: zero probability of error is only possible when $H(X|Y) = 0$.

The two-sided bound of theorem 9.1 is usually applied through either of two weaker, more convenient corollaries.

Corollary 9.3.

$$1 + P_e \log |\mathcal{X}| \;\geq\; H(X|Y)\,, \qquad \textit{equivalently} \qquad P_e \;\geq\; \frac{H(X|Y) - 1}{\log |\mathcal{X}|}.$$

Proof. $H(P_e) \leq 1$ for every $P_e \in [0,1]$ (theorem 5.7, applied to the two-point alphabet $\{0,1\}$), so the first inequality of theorem 9.1 gives $1 + P_e \log |\mathcal{X}| \geq H(P_e) + P_e \log |\mathcal{X}| \geq H(X|Y)$. Rearranging gives the second form. ∎

Corollary 9.4. *For any two random variables X, Y on the same alphabet, with $p = \mathbb{P}(X \neq Y)$,*

$$H(p) + p \log |\mathcal{X}| \;\geq\; H(X|Y)\,.$$

Proof. Apply theorem 9.1 with $\widehat{X} = Y$ (a valid, if trivial, estimator, and $X \to Y \to Y$ is trivially a Markov chain). ■

Corollary 9.5 (A sharper bound when $\widehat{X}$ takes values in $\mathcal{X}$)**.** *If $\widehat{X}$ is constrained to take values in $\mathcal{X}$ itself (rather than a possibly larger alphabet $\widehat{\mathcal{X}}$), then*

$$H(P_e) + P_e \log(|\mathcal{X}| - 1) \;\geq\; H(X|Y)\,.$$

Proof. Identical to the proof of theorem 9.1, except that, given $E = 1$ and $\widehat{X} = \hat{x} \in \mathcal{X}$, the true value X ranges over only the $|\mathcal{X}| - 1$ remaining symbols (it cannot equal $\hat{x}$, since $E = 1$ means $X \neq \widehat{X}$), so $H(X \mid \widehat{X}, E{=}1) \leq \log(|\mathcal{X}| - 1)$ replaces $\log|\mathcal{X}|$ throughout. ■

Example 9.6 (Fano's inequality is tight)**.** Suppose Y carries no information at all, so X must be guessed from its marginal law alone. Order the alphabet so $p_1 \geq p_2 \geq \cdots \geq p_m$ (where $m = |\mathcal{X}|$). The best guess is $\widehat{X} \equiv 1$, with $P_e = 1 - p_1$, and theorem 9.5 (with Y replaced by no information at all, so $H(X|Y)$ is simply $H(X)$) reads

$$H(P_e) + P_e \log(m - 1) \;\geq\; H(X).$$

This is tight: for the specific law $p = \big(1 - P_e, \frac{P_e}{m-1}, \ldots, \frac{P_e}{m-1}\big)$,

$$H(X) = H(P_e) + P_e \log(m - 1)$$

exactly, so no improvement of the constants in theorem 9.1 or theorem 9.5 is possible in general. For a concrete check, take $m = 4$ and $P_e = 0.6$, so $p = (0.4, 0.2, 0.2, 0.2)$: both sides equal 1.9219 bits.

10 Collision probability and entropy

Let X and X' be independent, identically distributed copies of a random variable with mass function p on $\mathcal{X}$. Their *collision probability* is

$$\mathbb{P}(X = X') = \sum_{x \in \mathcal{X}} p(x)^2,$$

the chance that two independent draws from the same source happen to agree. Intuitively, a more spread-out (higher-entropy) source should collide less often than a concentrated one. The next result makes this precise, with entropy as a lower bound on how rare collisions can be.

Lemma 10.1 (Collision probability bound)**.** *[Cover and Thomas, 2006, Lemma 2.10.1]. If X, X' are i.i.d. with entropy $H(X)$, then*

$$\mathbb{P}(X = X') \;\geq\; 2^{-H(X)},$$

with equality if and only if X is uniformly distributed on its alphabet.

Proof. Apply Jensen's inequality (theorem 5.2) to the strictly convex function $f(t) = 2^t$, under p:

$$2^{\mathbb{E}[\log p(X)]} \;\leq\; \mathbb{E}\big[2^{\log p(X)}\big].$$

The left side is $2^{\sum_x p(x) \log p(x)} = 2^{-H(X)}$. The right side is $\sum_x p(x)\, 2^{\log p(x)} = \sum_x p(x)^2 = \mathbb{P}(X = X')$, so $2^{-H(X)} \leq \mathbb{P}(X = X')$. Since $f(t) = 2^t$ is strictly convex, equality in Jensen's step holds iff $\log p(X)$ is almost surely constant, that is, iff $p(x)$ takes a single value on the support of X, which (since the probabilities sum to 1) means X is uniform on its alphabet. ■

Example 10.2 (Collision probability: a fair coin and a loaded die)**.** For a fair coin, $\mathbb{P}(X = X') = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}$, and $2^{-H(X)} = 2^{-1} = \frac{1}{2}$: equality, as theorem 10.1 predicts for a uniform source. For the loaded die of theorem 5.8, $p = \left(\frac{1}{2}, \frac{1}{10}, \ldots, \frac{1}{10}\right)$,

$$\mathbb{P}(X = X') = \left(\tfrac{1}{2}\right)^2 + 5\left(\tfrac{1}{10}\right)^2 = \tfrac{1}{4} + \tfrac{5}{100} = \tfrac{3}{10} = 0.3,$$

while $2^{-H(X)} = 2^{-2.161} \approx 0.224$, using $H(X) \approx 2.161$ bits from theorem 5.8. Indeed $0.3 > 0.224$: departing from uniformity, which lowered the entropy in theorem 5.8, raises the collision probability here, exactly the trade-off theorem 10.1 quantifies.

Corollary 10.3. *Let $X \sim p(x)$ and $X' \sim r(x)$ be independent, on the same alphabet $\mathcal{X}$. Then*

$$\mathbb{P}(X = X') \;\geq\; 2^{-H(p)-D(p\|r)}, \qquad \mathbb{P}(X = X') \;\geq\; 2^{-H(r)-D(r\|p)}.$$

Proof. By definition of relative entropy, $H(p) + D(p\|r) = -\sum_x p(x) \log r(x)$. Applying Jensen's inequality to $f(t) = 2^t$ under p, as in theorem 10.1, but with $\log r(X)$ in place of $\log p(X)$,

$$2^{-H(p)-D(p\|r)} = 2^{\sum_x p(x)\log r(x)} \leq \sum_x p(x)\, 2^{\log r(x)} = \sum_x p(x) r(x) = \mathbb{P}(X = X').$$

The second inequality follows by the symmetric argument, exchanging the roles of p and r. ∎

Remark 10.4 (this is one instance of a family). Theorem 10.1 is the case $\alpha = 2$ of a one-parameter family of entropies due to Rényi [1961]: for $\alpha > 0$, $\alpha \neq 1$, the *Rényi entropy of order α* is

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \sum_x p(x)^\alpha,$$

which recovers Shannon entropy as $\alpha \to 1$ (by l'Hôpital's rule) and satisfies $H_2(X) = -\log \mathbb{P}(X = X')$, the *collision entropy*. Rényi entropy is a non-increasing function of α, so in particular $H_2(X) \leq H_1(X) = H(X)$, which is exactly $-\log \mathbb{P}(X = X') \leq H(X)$, that is, theorem 10.1. Seen this way, the collision bound is not an isolated trick with the function 2^t, it is the $\alpha = 1$-versus-$\alpha = 2$ instance of a monotonicity property that holds for the whole family, of which Shannon entropy ($\alpha \to 1$) and the maximum-entropy bound of theorem 5.7 ($\alpha \to 0$, where $H_0(X) = \log|\{x : p(x) > 0\}|$) are the two most commonly used other members.

11 Summary

One-page summary

Entropy. $H(X) = -\sum_x p(x) \log p(x) = \mathbb{E} \log \frac{1}{p(X)}$ (theorem 1.1). $H(X) \geq 0$, and $H(X) \leq \log|\mathcal{X}|$ with equality iff X is uniform (theorems 1.3 and 5.7). $H(p)$ is concave in p (theorem 6.5).

Joint and conditional entropy. $H(X, Y) = H(X) + H(Y|X)$ (theorem 2.3), and more generally $H(X^n) = \sum_i H\left(X_i \middle| X^{i-1}\right)$ (theorem 2.8). $H(X|Y) \leq H(X)$ on average, with equality iff $X \perp\!\!\!\perp Y$ (theorem 5.9).

Relative entropy and mutual information. $D(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)} \geq 0$, with equality iff $p = q$ (theorem 5.3). $I(X;Y) = D(p_{XY}\|p_X p_Y) = H(X) - H(X|Y) = H(X) + H(Y) - H(X,Y) \geq 0$, with equality iff $X \perp\!\!\!\perp Y$ (theorems 3.5, 3.6 and 5.6).

Chain rules. $I(X^n;Y) = \sum_i I\big(X_i;Y\big|X^{i-1}\big)$ (theorem 4.2). $D(p_{XY}\|q_{XY}) = D(p_X\|q_X) + D(p_{Y|X}\|q_{Y|X} \mid p_X)$ (theorem 4.4).
Log-sum inequality and convexity. $\sum_i a_i \log \frac{a_i}{b_i} \geq (\sum_i a_i) \log \frac{\sum_i a_i}{\sum_i b_i}$, equality iff a_i/b_i is constant (theorem 6.1). $D(p\|q)$ is jointly convex in (p,q) (theorem 6.4). $I(X;Y)$ is concave in p_X for fixed channel, convex in the channel for fixed p_X (theorem 6.7).
Data processing. If $X \to Y \to Z$, then $I(X;Y) \geq I(X;Z)$ (theorem 7.2), with equality iff $X \to Z \to Y$ also holds. Functions of the data cannot create information (theorem 7.3).
Sufficiency. $T(X)$ is sufficient for θ iff $I(\theta;T(X)) = I(\theta;X)$ for every prior on θ (theorem 8.2), equivalently iff $f_\theta(x) = g(T(x),\theta)h(x)$ (theorem 8.5).
Fano's inequality. For $\widehat{X} = g(Y)$ with $X \to Y \to \widehat{X}$ and $P_e = \mathbb{P}(\widehat{X} \neq X)$,

$$H(P_e) + P_e \log|\mathcal{X}| \;\geq\; H(X \mid \widehat{X}) \;\geq\; H(X|Y)$$

(theorem 9.1), sharp, and usable in the weaker form $P_e \geq (H(X|Y) - 1)/\log|\mathcal{X}|$ (theorem 9.3).
Collision probability. For i.i.d. X, X', $\mathbb{P}(X = X') \geq 2^{-H(X)}$, equality iff X is uniform (theorem 10.1), the $\alpha = 2$ case of the monotonicity of Rényi entropy H_α in α.

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